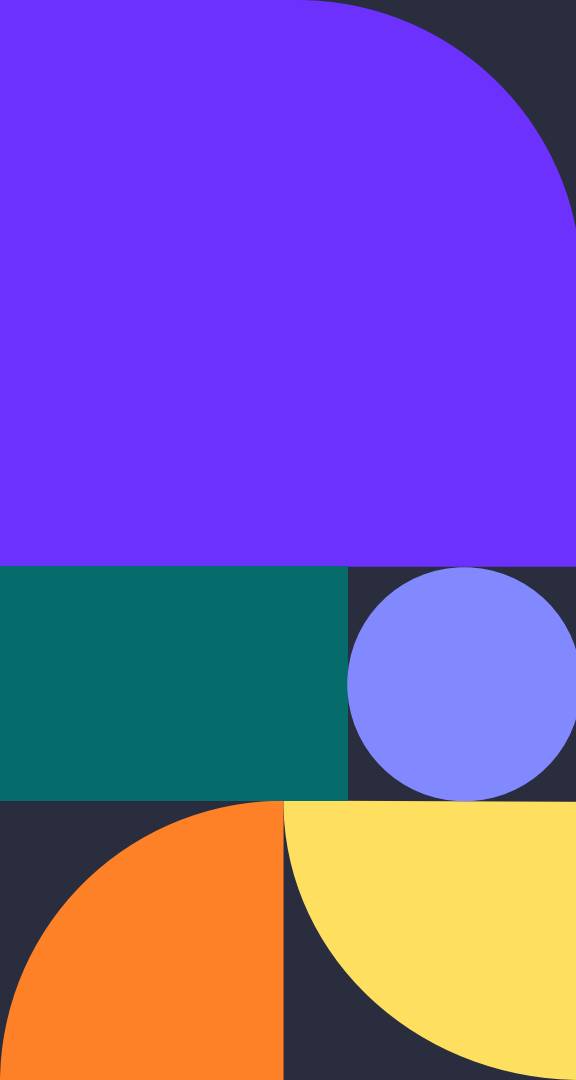




# Project B – Places Attribute Conflation

By: James Conrad Manlangit

Sponsored By: Overture Maps  
Foundation

# Contents

**1. Project Overview**

**2. Approach &  
Methodology**

**3. Demo**

**4. OKRs & Outcomes**

**5. Reflection**



# Project Overview:

- Overture Maps gathers location and business information from multiple different sources.
  - Issue: Sources can conflict = Differences in addresses, business names, etc.
  - Which attributes do we select?



Correct Address = ?



Source A's Address:

4202 E Fowler Ave,  
Tampa, FL,  
33620-0007



Source B's Address:

USF Library, Tampa,  
FL, 33620

# Contents

**1. Project Overview**

**2. Approach &  
Methodology**

**3. Demo**

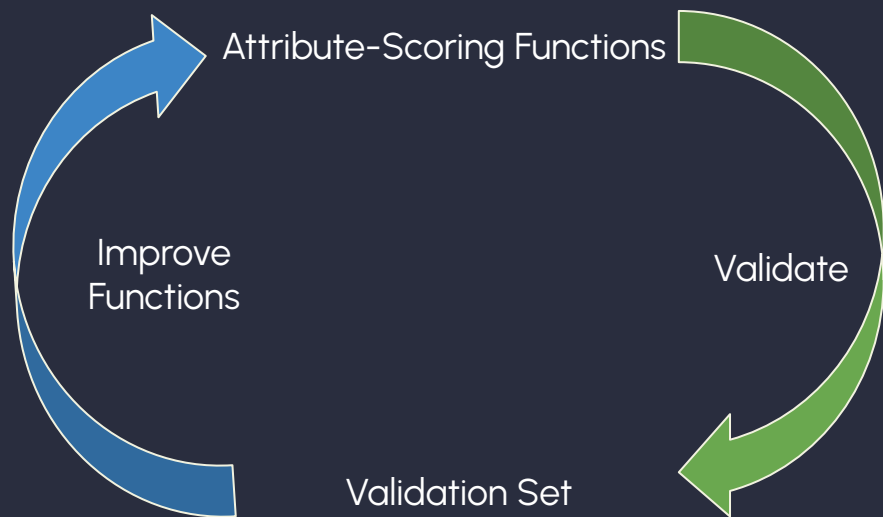
**4. OKRs & Outcomes**

**5. Reflection**



# Rule-Based Approach

- Built functions scoring each major location attribute a quality score.
  - Compare quality score between Source A and Source B.
  - Build "golden dataset"
- Validate - Matched Project B dataset locations with Yelp Open Dataset
  - Repeatedly use label disagreements to improve scoring functions.



# Contents

**1. Project Overview**

**2. Approach &  
Methodology**

**3. Demo**

**4. OKRs & Outcomes**

**5. Reflection**



# Labeling

## Source A

| base_addresses                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------|
| [{"freeform": "7643 Gate Pkwy Ste 104", "locality": "Jacksonville", "region": "FL", "country": "US", "postcode": "32256"}] |
| [{"freeform": "1158 N 2nd St", "locality": "El Cajon", "region": "CA", "country": "US", "postcode": "92021"}]              |
| [{"freeform": "49 Emnotweni Avenue", "locality": "Mbombela", "region": "", "country": "ZA", "postcode": "1201"}]           |
| [{"freeform": "875 Mansell Rd Ste B1", "locality": "Roswell", "region": "GA", "country": "US", "postcode": "30076"}]       |
| [{"freeform": "Calle Rei Jaume I 9", "locality": "Alicante", "region": "Alacant", "country": "ES", "postcode": "03509"}]   |
| [{"freeform": "43821 Shady Creek", "locality": "Temecula", "region": "CA", "country": "US", "postcode": "92590"}]          |
| [{"freeform": "6254 Duke St", "locality": "Alexandria", "region": "VA", "country": "US", "postcode": "22312"}]             |

## Source B

| addresses                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------|
| [{"freeform": "7643 Gate Pkwy", "locality": "Jacksonville", "postcode": "32256-2892", "region": "FL", "country": "US"}]                   |
| [{"freeform": "1158 N 2nd St", "locality": "El Cajon", "postcode": "92021-5023", "region": "CA", "country": "US"}]                        |
| [{"freeform": "Mazda Nelspruit, 49 Emnotweni Avenue", "locality": "Riverside Park", "postcode": "1201", "region": "MP", "country": "ZA"}] |
| [{"freeform": "875 Mansell Rd B1", "locality": "Roswell", "region": "GA", "country": "US", "postcode": "30076"}]                          |
| [{"freeform": "Carrer del Rei Jaume I 9", "locality": "Finestrat", "postcode": "03509", "country": "ES"}]                                 |
| [{"freeform": "43821 Shady Crk", "locality": "Temecula", "region": "CA", "country": "US", "postcode": "92590"}]                           |
| [{"freeform": "6254 Duke St", "locality": "Alexandria", "postcode": "22312", "region": "VA", "country": "US"}]                            |

## Source A

| quality_score_addresses_base |
|------------------------------|
| 1                            |
| 1                            |
| 0.75                         |
| 1                            |
| 0.5                          |
| 0.9500000000000001           |
| 1                            |

## Source B

| quality_score_addresses |
|-------------------------|
| 0.7500000000000001      |
| 0.8                     |
| 0.4500000000000007      |
| 1                       |
| 0.3999999999999997      |
| 0.9500000000000001      |
| 1                       |

Scoring



Labeling



| label_addresses |
|-----------------|
| 1               |
| 1               |
| 1               |
| 2               |
| 1               |
| 2               |
| 2               |

| addresses                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------|
| [{"freeform": "7643 Gate Pkwy Ste 104", "locality": "Jacksonville", "region": "FL", "country": "US", "postcode": "32256"}] |
| [{"freeform": "1158 N 2nd St", "locality": "El Cajon", "region": "CA", "country": "US", "postcode": "92021"}]              |
| [{"freeform": "49 Emnotweni Avenue", "locality": "Mbombela", "region": "", "country": "ZA", "postcode": "1201"}]           |
| [{"freeform": "875 Mansell Rd B1", "locality": "Roswell", "region": "GA", "country": "US", "postcode": "30076"}]           |
| [{"freeform": "Calle Rei Jaume I 9", "locality": "Alicante", "region": "Alacant", "country": "ES", "postcode": "03509"}]   |
| [{"freeform": "43821 Shady Crk", "locality": "Temecula", "region": "CA", "country": "US", "postcode": "92590"}]            |
| [{"freeform": "6254 Duke St", "locality": "Alexandria", "postcode": "22312", "region": "VA", "country": "US"}]             |

Golden Dataset

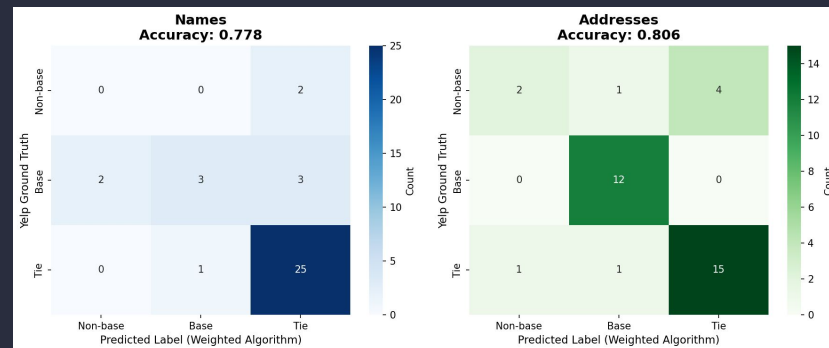
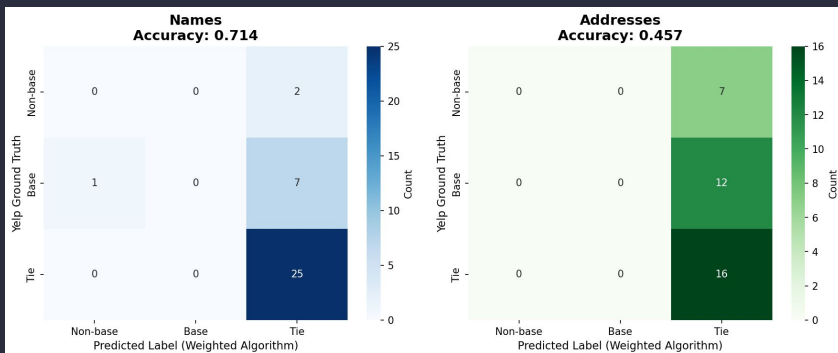


# Validation

# Fuzzy String Matching with Yelp Dataset



| base_address | base_addr_sim | address_string   | addr_sim | yelp_address | label_addresses | yelp_truth_addresses |
|--------------|---------------|------------------|----------|--------------|-----------------|----------------------|
| 8110 N Colle | 1             | 8110 N College A | 0.94     | 8110 N Colle | 1               | 1                    |
| 7041 N Cami  | 0.85          | 7041 N Camino M  | 0.94     | 7020 N Cami  | 2               | 0                    |



# Contents

**1. Project Overview**

**2. Approach &  
Methodology**

**3. Demo**

**4. OKRs & Outcomes**

**5. Reflection**



# Final OKRs

✅ = Met

⚠️ = Partially Met

- Objective 1: Build a high-quality labeled ground-truth dataset from pre-matched places data ✅
  - KR1: Achieve  $\geq 95$  % non-null coverage across core label attributes (names, addresses, phones, categories) ✅
- Objective 2: Develop a reliable and interpretable selection algorithm for attribute conflation ⚠️
  - K1: Achieve  $\geq 80$  % overall attribute-level accuracy when comparing conflation outputs against the labeled ground truth. ⚠️ (Achieved 80.6% address accuracy, but only 77.8% name accuracy).
  - KR2: Validate  $\geq 85$ % attribute-specific format correctness for URLs and addresses (valid URLs, complete postal addresses). ✅
- Objective 3: Produce a comprehensive performance evaluation report ⚠️
  - KR1: Evaluate pipeline accuracy and precision per attribute on validation set ( $\geq 1000$  records) ⚠️
  - KR2: Generate 1-2 performance visualizations and include them in a single finalized evaluation report. ✅

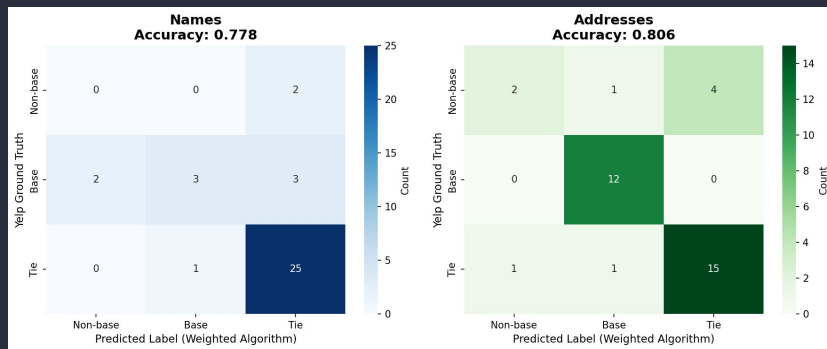
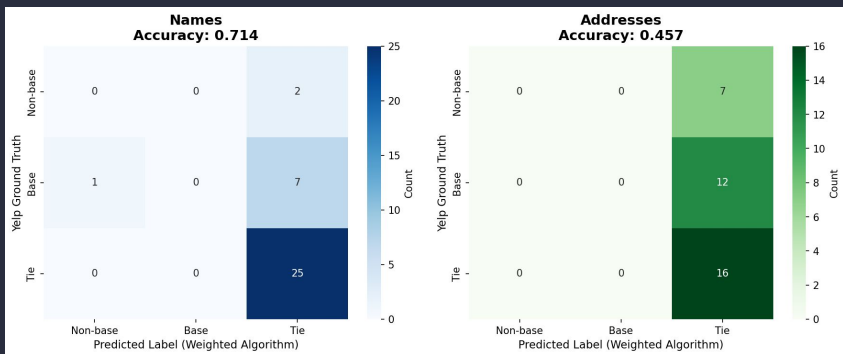
# Results

What I accomplished:

- Scalable baseline algorithm that can select more reliable attributes between conflicting sources.

Areas of Improvement:

- Validated and improved upon only using Yelp Dataset -> Focus only on US/Canada
- Linear solution to a non-linear problem
  - Not all ties are perfect



# Contents

**1. Project Overview**

**2. Approach &  
Methodology**

**3. Demo**

**4. OKRs & Outcomes**

**5. Reflection**



# Reflection:

If I had one more quarter...

- Research and try to implement more ML-based solutions.
- Use a more global ground-truth proxy for validation
- Research and implement a less expensive ground-truth proxy matching method.

**Thank You!**